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Stochastic Modelling of Liquidity Provider Returns in Uniswap

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Abstract
Current work introduces a theoretically grounded framework for estimating returns of liquidity providers (LPs) in automated market makers (AMM), with a focus on Uniswap V3 (Concentrated Liquidity AMM) and Uniswap V2 (Constant Product AMM). We explicitly model returns as the sum of fee income and impermanent loss, deriving theoretical mathematical expectations for both components under Geometric Brownian Motion price dynamics. While existing models often estimate fees as a simple function of historical trading volume or arbitrage dynamics entirely, our approach bridges market microstructure theory and decentralised finance mechanics to provide a practical valuable tool. The proposed framework decomposes fee income into ordinary user transactions and arbitrage flows, explicitly accounting for position activation probability within user-defined price ranges under stochastic price dynamics. To address real-world complexities – such as variable swap sizes and fluctuating pool liquidity – we introduce interpretable correction factors that adapt to market conditions. Validating through extensive Monte-Carlo simulations with realistic market microstructure, our model demonstrates robustness across diverse volatility regimes and pool configurations. For practitioners, the proposed framework enables LPs to optimise price ranges, compare fee tiers and predict returns under different market conditions. For researchers, it establishes a foundational methodology for analysing liquidity provision in AMMs that connects theoretical price dynamics to impermanent loss and fee generation. We further incorporate operational costs (blockchain gas fees) into the profitability analysis, demonstrating how transaction costs shift optimal position widths and rebalancing thresholds. The calibration constants governing the model are validated through regime-dependent estimation on historical on-chain data and out-of-sample testing, establishing measurable robustness. This work advances the understanding of sustainable liquidity provision in decentralised exchanges and offers practical recommendations for rational capital allocation in DeFi protocols.

Keywords: *Automated Market Makers (AMMs), Concentrated Liquidity, Uniswap V3, Liquidity Provision, Impermanent Loss, Stochastic Price Dynamics, Arbitrage Dynamics*

JEL Classifications: *D40, D47, D53, D01, D02*

1. Introduction

With the rise of blockchain networks, a new era in decentralised financial technologies has emerged. The evolution of the smart contract concept [1] enabled the creation of autonomous decentralised financial protocols. One of the breakthrough concepts of DeFi is automated market makers (AMMs) [2] – an algorithmic liquidity mechanism based on predefined mathematical invariants, which is replacing traditional order books. Early AMM implementations, such as Uniswap V2 [3] (CPMM), employed a simple and robust constant product invariant for pricing, but suffered from a fundamental limitation: liquidity was supplied uniformly across the entire price range, resulting in significant capital inefficiency [4]. Thereby, Uniswap V3 [5] (CLMM) with its innovative approach of concentrated liquidity allows liquidity providers (LPs) to allocate their capital within user-specified price intervals, thereby amplifying fee income but exposing potential risk of investments. Therefore, managing of concentrated liquidity provision is a very challenging task: around 49.5% of LPs in Uniswap V3 have negative investment returns [6]. Consequently, the value of a concentrated liquidity position critically depends on the protocol's fee structure, the transactional activity generating swap fees, and the underlying price dynamics and volatility. Under these conditions, understanding the relationship between expected returns and market parameters becomes crucial both for market participants and protocol designers.

Research on concentrated liquidity has become a central theme in the decentralised finance literature, while the theoretical understanding of liquidity provision in CLMMs remains fragmented. A recent comprehensive survey [7] has synthesised various existing descriptions of concentrated liquidity mechanisms from major protocols, providing theoretical unification for the literature. In [8], a deterministic analysis of impermanent loss at a fixed price is given. The impact of position duration on revenue is considered in [4]; a more extended version including other important features is presented in [9]. A stochastic approach is described in [10] where a theoretical model for accumulated fees is derived assuming Geometric Brownian Motion for price dynamics, which is further extended in [11] to estimate optimal investment horizons and maximal returns. Closed mathematical analysis in [12] describes another aspect of dynamics – strategic behaviour of arbitrageurs. Considerable publication attention has been devoted to designing optimal dynamic algorithms for managing liquidity positions [13, 14]. The formal concept of JIT-liquidity, which allows LPs to earn instantly while providing liquidity for a single large swap, is described in [15]. Another important research direction concerns delta-neutral strategies to hedge impermanent loss risks using market derivatives [16, 17, 18]. In work [19], a novel primitive for rebalancing Uniswap V3 liquidity positions is introduced, enabling reinvestment of fees and reallocation of assets without requiring swaps on external markets.

Crucially, two interconnected gaps persist: (i) no rigorous model unifies price dynamics, arbitrage mechanics, and ordinary swap arrivals to predict expected fee income under realistic market conditions; (ii) not model captures the relationship between impermanent loss and fee revenue while accounting for volatility regimes, fee tiers and position width. This paper bridges these gaps by developing a stochastic valuation model for concentrated liquidity positions that integrates continuous price movements with discrete trade arrivals and arbitrage dynamics heuristics.

We make three core contributions. First, we derive a closed-form expression for the expected dimensionless impermanent loss of a Uniswap V3 position under Geometric Brownian Motion (GBM) price dynamics (Theorem 1), generalising prior deterministic results to a probabilistic setting. This reveals how impermanent loss (IL) scales with volatility, time horizon and position width – an absent insight from the current literature. Second, we formulate an analytically tractable heuristic model for expected fee income that decomposes revenue into ordinary user swaps and arbitrage-driven swaps, with a calibration constant for market volatility and fee tiers (Theorem 2). Our model uniquely incorporates the variance-balancing mechanism that governs arbitrage frequency, linking fee structure to price deviation dynamics. Third, we analyse the conditions for profitability by unifying IL and fee expectations, providing an optimisation framework for LPs that adapts to volatility regimes. Fourth, we extend the profitability analysis to incorporate blockchain operational costs (gas fees), showing how transaction costs shift optimal position widths and create minimum viable position sizes for each strategy. These theoretical results are validated through Monte-Carlo simulations and backtested against real on-chain data on a popular Uniswap V3 pool, with explicit calibration methodology and out-of-sample robustness checks for the model's correction factors.

The paper is structured as follows. Section 1 provides a brief introduction and a literature review of relevant works. Section 2 establishes the theoretical preliminaries: we formalise Uniswap V2/V3, metrics for IL and arbitrage constraints. Section 3 presents the core of the proposed stochastic valuation framework, including a calibration and validation methodology for the model's correction factors. Section 4 details empirical validation of dynamic strategies using real-world data, incorporating operational cost analysis. Finally, we conclude with key limitations, practical recommendations and promising directions for future research.

2. Theoretical Background and Preliminaries

2.1. Constant Product Market Maker

Uniswap V2 implements a constant product market-making model (CPMM), where the product of the reserves of two assets x and y is held constant:

$$x \cdot y = L^2,$$

for liquidity $L > 0$. The spot price p (in units of y per x) is derived from the marginal exchange rate

$$p = \frac{\partial y}{\partial x}\Big|_{L=const} = \frac{y}{x} = \frac{L^2}{x^2}.$$

A LP depositing assets $(\Delta x, \Delta y)$ receives liquidity tokens proportional to the LP's share of the pool.

The position value function in terms of asset y:

$$V_y(p) = p \cdot x + y = 2L\sqrt{p} \tag{1}$$

This model suffers from capital inefficiency: liquidity is uniformly distributed across all prices, leading to low capital utilisation.

2.2. Constant Liquidity Market Maker

Uniswap V3 introduces a concentrated liquidity market-making model (CLMM), allowing LPs to allocate capital within custom price ranges $[p_l, p_u]$. This innovation addresses the capital inefficiency of Uniswap V2 by enabling LPs to concentrate liquidity around expected price ranges, achieving higher capital efficiency for narrow ranges.

For a position with price range $[p_l, p_u]$, the current liquidity L and price p, the amounts of assets x and y are given by:

$$x(p) = L \begin{cases} \dfrac{1}{\sqrt{p_l}} - \dfrac{1}{\sqrt{p_u}}, & if\ p < p_l \\ \dfrac{1}{\sqrt{p}} - \dfrac{1}{\sqrt{p_u}}, & if\ p_l \le p \le p_u \\ 0, & if\ p > p_u \end{cases} \tag{2}$$

$$y(p) = L \begin{cases} 0, & if\ p < p_l \\ \sqrt{p} - \sqrt{p_l}, & if\ p_l \le p \le p_u \\ \sqrt{p_u} - \sqrt{p_l}, & if\ p > p_u \end{cases} \tag{3}$$

We say the position is active if $p_l \le \mathrm{p} \le p_u$, otherwise inactive.

Using (2) and (3), one can derive Uniswap V3 position invariant [5]:

$$\underbrace{\left(x + \frac{L}{\sqrt{\mathrm{p_u}}}\right)}_{x_{virtual}} \cdot \underbrace{\left(y + L\sqrt{p_l}\right)}_{y_{virtual}} = L^2$$

which generalises the Uniswap V2's constant product for virtual amounts $(x_{virtual}, y_{virtual})$. A visualisation of this position is shown in Figure 1. Note that in the active price interval $[p_l, p_u]$, the CLMM invariant exactly coincides with that of CPMM.

The value function in terms of asset y of Uniswap v3 position with liquidity L, price bounds $[q_l, q_u]$, and current price p_t is:

$$\mathrm{V_y(p_t)} = p_t \cdot x(p_t) + y(p_t) = L \begin{cases} \frac{\mathrm{p_t}}{\sqrt{p_l}} - \frac{\mathrm{p_t}}{\sqrt{p_u}}, & if\ p < p_l \\ 2\sqrt{\mathrm{p_t}} - \frac{\mathrm{p_t}}{\sqrt{p_u}} - \sqrt{p_l}, & if\ p_l \le p \le p_u \\ \sqrt{p_u} - \sqrt{p_l}, & if\ p > p_u \end{cases} \tag{4}$$

which is continuous at $p = p_l$ and $p = p_u$. Note that in the limiting case of an unbounded position ($p_l \to 0$, $p_u \to \infty$), Uniswap V3 position value (4) reduces to Uniswap V2 (1). However, it should be considered that, in practical implementations, Uniswap V2 fees are automatically reinvested into the pool as additional liquidity, which is described in detail in [20]. In contrast, Uniswap V3 fees are accrued to a separate account and do not contribute to pool liquidity. Nevertheless, for analytical tractability in the Uniswap V2 case, we will neglect this distinction in approximations over small position lifetime.

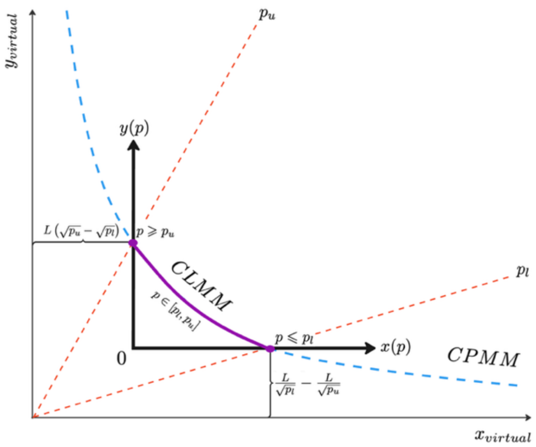

Figure 1. The visualisation of the concentrated liquidity invariant under the introduced notation.

For further derivations, it is convenient to consider a dimensionless position value, normalised by $2L\sqrt{p_0}$ – the value of a Uniswap V2 position with liquidity L:

$$\upsilon_y(\mathrm{p_t}) = \frac{1}{2}\frac{\mathrm{V_y(p_t)}}{\mathrm{L}\sqrt{\mathrm{p_0}}} = \frac{1}{2}\begin{cases} \frac{p_t}{\sqrt{p_0 \cdot p_l}} - \frac{p_t}{\sqrt{p_0 \cdot p_u}}, & if\ p < p_l \\ 2 \cdot \sqrt{\frac{p_t}{p_0}} - \frac{p_t}{\sqrt{p_0 \cdot p_u}} - \sqrt{\frac{\mathrm{p_l}}{\mathrm{p_0}}}, & if\ p_l \le p \le p_u \\ \sqrt{\frac{p_u}{p_0}} - \sqrt{\frac{p_l}{p_0}}, & if\ p > p_u \end{cases} \tag{5}$$

2.3. Impermanent Loss

Impermanent loss represents the opportunity cost incurred by LPs when the relative price of assets in a pool changes after their deposit [4]. This loss arises from the automated market maker's rebalancing mechanism, which executes implicit trades against LPs as prices shift. Crucially, IL is impermanent, because it materialises only when liquidity is withdrawn at a price different from the deposit price. If prices revert to their initial state before withdrawal, the loss disappears.

The absolute IL (in units of asset y) is defined as the difference between the liquidity provision strategy and passive holding outside the pool. For an initially active Uniswap V3 concentrated position, it can be derived as

$$\mathrm{IL_{abs}}(p_t) = V_y(p_t) - V_{\mathrm{hold}}(p_t) = V_y(p_t) - (x_0 \cdot p_t + y_0)$$

To enable cross-pool and cross-asset comparisons, we normalise absolute IL by the scale $2L\sqrt{p_0}$ [8]:

$$\mathrm{IL_{norm}}(p_t) = \frac{\mathrm{IL_{abs}}(p_t)}{2L\sqrt{p_0}} = \upsilon_y(p_t) - \frac{x_0 \cdot p_t + y_0}{2L\sqrt{p_0}}.$$

When we assume the position activity at $t = 0$, $\mathrm{IL_{norm}}(p_t)$ is derived as

$$\mathrm{IL_{norm}}(p_t) = \upsilon_y(p_t) - \frac{1}{2}\left(1 + \frac{p_t}{p_0} - \frac{p_t}{\sqrt{p_0 \cdot p_u}} - \sqrt{\frac{p_l}{p_0}}\right). \tag{6}$$

For the case $p_l \le p_t \le p_u$, the expression is simplified:

$$\mathrm{IL_{norm}}(p_t) = \sqrt{\frac{p_t}{p_0}} - \frac{1}{2} - \frac{1}{2}\frac{p_t}{p_0} = -\frac{1}{2}\left(\sqrt{\frac{p_t}{p_0}} - 1\right)^2,$$

from which it follows that $IL < 0$ for all $p_t \ne p_0$. Moreover, note that $\mathrm{IL_{norm}}$ coincides with the expression for Uniswap V2 [8] and does not explicitly depend on the range bounds, which provides us with a way to reduce a position Uniswap V3 to Uniswap V2. In contrast, the range bounds p_l and p_u appear in $\mathrm{IL_{abs}}$ because the liquidity amount L depends on p_l and p_u (4). For $p_l \le p_{0,t} \le p_u$ and initial position value V_0 it is derived from (3) and (4):

$$\mathrm{IL_{abs}}(p_t) = -\frac{\left(\sqrt{\frac{p_t}{p_0}} - 1\right)^2}{2 - \sqrt{\frac{p_0}{p_u}} - \sqrt{\frac{p_l}{p_0}}} \cdot V_0. \tag{7}$$

The dependence of $\mathrm{IL_{abs}}$ on different position price ranges is shown in Figure 2. A wider price range leads to greater IL when the price deviates from the deposit price. For a position of infinite width (i.e., Uniswap V2), the IL is minimal for any fixed target price p_t, which follows directly from (7).

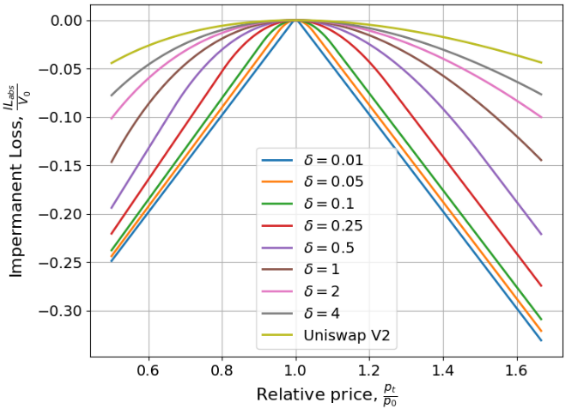

Figure 2. The dependence of IL on the width of a geometrically symmetric position defined by $\frac{p_u}{p_0} = \frac{p_0}{p_l} = \delta$ as a function of the half-width parameter δ.

2.4. Dynamics of Geometric Brownian Motion

Geometric Brownian Motion is a fundamental stochastic process widely used in mathematical finance to model asset prices [21]. It is characterised by the following stochastic differential equation (SDE):

$$dp_t = \mu p_t dt + \sigma p_t dW_t, \tag{8}$$

where p_t represents the price process at time t, μ is the drift coefficient (expected return), σ is the volatility parameter, and W_t is a standard Wiener process (Brownian motion).

The analytical solution [21] for the price process p_t with initial condition p_0 is given by:

$$p_t = p_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right).$$

In other words, the stochastic price dynamics define a log-normal random variable with the following parameters

$$p_t \sim \text{LogNormal}\left(\ln p_0 + \left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right). \tag{9}$$

The Monte-Carlo simulation of the GBM dynamics is visualised in Figure 3. It is evident that, as the horizon t increases, the probability density of p_t becomes increasingly diffused in accordance with the exact analytical solution (its variance grows). Next, we will apply a Monte-Carlo procedure to simulate price paths and validate the theoretical results.

Note that p_t models the external market price (e.g., on a centralised exchange, CEX). The internal price p_{AMM} or state price on AMM (DEX) often lags behind the CEX price, resulting in a non-zero difference. This difference can create an arbitrage opportunity, the mechanics of which are considered in the next section.

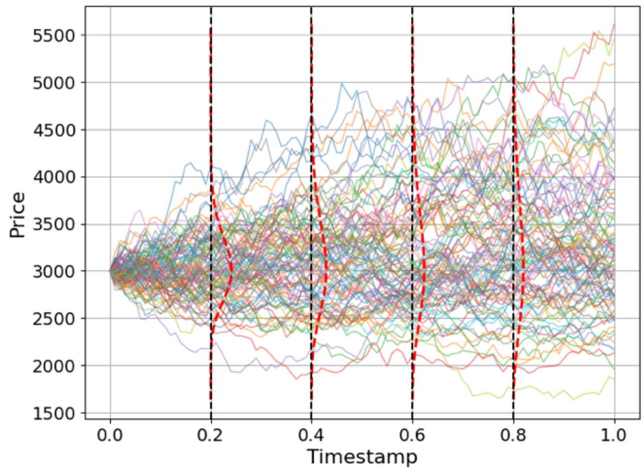

Figure 3. The dynamics of GBM with $\mu = 0.1$, and $\sigma = 0.25$, $p_0 = 3000$. The simulation consists of 100 sample paths. The red dashed line shows the density function of the random variable p_t (9).

2.5. Arbitrage Opportunity with Infinite Liquidity Market

A core mechanism aligning AMM prices with external markets is arbitrage. We consider two markets for the same asset pair: an external reference market with a "true" (or "fair") price p_t (8), assumed to have infinite liquidity, and a decentralised AMM pool with an internal price $p_{\text{AMM}}(t)$. An arbitrageur can execute a risk-free profit by buying in the cheaper market and selling in the more expensive one. However, this trade incurs costs: the AMM's fee ϕ and execution costs $\epsilon > 0$, which captures the gas expenditure required to submit the arbitrage transaction on the blockchain. Specifically, for an arbitrage swap of volume V with gas cost G_{gas} denominated in the numeraire, the effective cost parameter is $\epsilon \approx G_{gas}/V$. On Ethereum mainnet during the study period (Oct 2024–Oct 2025), typical gas costs for a single swap ranged from approximately 1-40$ depending on network congestion, while on Layer-2 networks (e.g., Arbitrum, Base) costs were 10-100× lower. This cost asymmetry becomes material in Section 4, where we analyse rebalancing strategies that require multiple on-chain transactions per cycle.

Without loss of generality, we assume $p_{\text{AMM}}(t) > p_t$. The arbitrageur buys Δx at p_t externally and sells in the AMM at an effective price of $p_{\text{AMM}}(t)(1-\phi)$ to get $\Delta x^* > \Delta x$. For the existence of the profitable trade, the effective AMM price must exceed the external cost: $p_{\text{AMM}}(t)(1-\phi) > p_t(1+\epsilon)$. For small ϕ and ϵ, this simplifies to $\frac{p_{\text{AMM}}(t)}{p_t} > 1 + \phi + \epsilon$. A symmetric case for $p_{\text{AMM}}(t) < p_t$ yields the condition $\frac{p_{\text{AMM}}(t)}{p_t} < 1 - \phi - \epsilon$. Thus, an arbitrage opportunity exists if and only if the relative price deviation exceeds a threshold

$$\left|\frac{p_{\text{AMM}}(t)}{p_t} - 1\right| > \phi + \epsilon, \tag{10}$$

which defines the no-arbitrage band $[1-\phi-\epsilon, 1+\phi+\epsilon]$. The band's width, proportional to ϕ, is crucial: it links the pool's fee structure to the stochastic dynamics of $p_{\text{AMM}}(t)$, forming the basis for our fee income model.

3. Stochastic Valuation of Concentrated Liquidity Positions

The valuation of concentrated liquidity positions in Uniswap V3 requires accounting for the stochastic evolution of asset prices and discrete trade arrivals. This section develops a framework for position valuation under these dynamics, integrating continuous price diffusion with event-driven execution.

3.1. Expected Losses Under Geometric Brownian Motion

We first establish a foundational result for the expected dimensionless position value of concentrated liquidity (5) under continuous price dynamics.

Theorem 1. At time $t > 0$, the expected dimensionless value $E[\upsilon_y(t)]$ of a concentrated liquidity position with liquidity L in range $[p_l,\ p_u]$ under GBM price dynamics (8) is

$$
\begin{aligned}
E[\upsilon_y(t)] = {} & \left(\sqrt{\frac{p_0}{p_l}} - \sqrt{\frac{p_0}{p_u}}\right)\frac{e^{\mu t}}{2}\Phi\left(\frac{\ln\frac{p_l}{p_0} - \left(\mu + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right) + \\
& + e^{\left(\frac{\mu}{2} - \frac{\sigma^2}{8}\right)t}\left[\Phi\left(\frac{\ln\frac{p_u}{p_0} - \mu t}{\sigma\sqrt{t}}\right) - \Phi\left(\frac{\ln\frac{p_l}{p_0} - \mu t}{\sigma\sqrt{t}}\right)\right] - \\
& - \frac{1}{2}\sqrt{\frac{p_l}{p_0}}\left[\Phi\left(\frac{\ln\frac{p_u}{p_0} - \left(\mu - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right) - \Phi\left(\frac{\ln\frac{p_l}{p_0} - \left(\mu - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right)\right] - \\
& - \sqrt{\frac{p_0}{p_u}}\frac{e^{\mu t}}{2}\left[\Phi\left(\frac{\ln\frac{p_u}{p_0} - \left(\mu + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right) - \Phi\left(\frac{\ln\frac{p_l}{p_0} - \left(\mu + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right)\right] + \\
& + \frac{1}{2}\left(\sqrt{\frac{p_u}{p_0}} - \sqrt{\frac{p_l}{p_0}}\right)\left[1 - \Phi\left(\frac{\ln\frac{p_u}{p_0} - \left(\mu - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right)\right],
\end{aligned}
$$

where $\Phi(\cdot)$ is the CDF of standard normal distribution [22].

Proof. See Appendix A.

The proposed Theorem 1 can be used to derive an estimate for the value of a CPMM position by considering the CLMM in its limiting case as follows.

Corollary 1. In the case of infinite position width ($p_l \to 0$ and $p_u \to \infty$) the expected dimensionless value of the liquidity position exhibits the following asymptotic behaviour

$$E[\upsilon_y(t)] \to e^{\left(\frac{\mu}{2}-\frac{\sigma^2}{8}\right)t},$$

or

$$E[V_y(p_t)] \to 2L\sqrt{p_0}e^{\left(\frac{\mu}{2}-\frac{\sigma^2}{8}\right)t},$$

which fully coincides with the expression from [23].

We note that the results of Theorem 1 are novel compared to the previous works [4, 8, 18], which derive IL for one final price. In our case, under the assumption that p_t evolves according to continuous GBM dynamics, we can establish a foundational result for the expected dimensionless impermanent loss $\text{IL}_{norm}(p_t)$ to estimate on average.

Corollary 2. For an initially active position (at time $t = 0$), the expectation of normalised IL is given by

$$E[IL_{norm}(p_t)] = E[\upsilon_y(t)] - \frac{1}{2}\left(1 - \sqrt{\frac{p_0}{p_u}}\right)e^{\mu t} - \frac{1}{2}\left(1 - \sqrt{\frac{p_l}{p_0}}\right).$$

Proof. Using the linearity of expectation for (6) and the fact that, under GBM dynamics,

$$E[p_t] = p_0 e^{\mu t},$$

we recover the original expression.

Corollary 3. The expectation of absolute IL with initially active position ($p_l \le p_0 \le p_u$) and value V_0 is

$$E[IL_{abs}(p_t)] = 2L\sqrt{p_0}\cdot E[IL_{norm}(p_t)] = \frac{2V_0 \cdot E[IL_{norm}(p_t)]}{2 - \sqrt{\frac{p_0}{p_u}} - \sqrt{\frac{p_l}{p_0}}}.$$

The mathematical expectation of absolute IL is shown in Figure 4 for different position widths. As a verification, the 95% confidence interval calculated using Monte-Carlo simulations were provided. The observed behaviour is fully consistent with Figure 2, which shows that the narrower the position, the larger the IL. The lowest expected losses are observed in the case of Uniswap V2.

Similarly, Figure 5 shows the effect of the volatility parameter σ in the zero-drift case $\mu = 0$. It is evident that the higher the volatility σ, the greater the expected IL. This is why, in practice, it is generally recommended to widen the liquidity range when trading in high volatility environments.

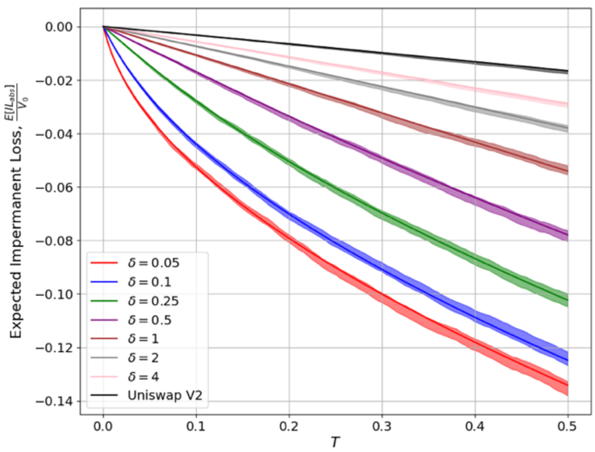

Figure 4. Expectation of IL as dependence of the investment horizon T for geometrically symmetric positions with different half-widths δ. The filled area represents the 95% confidence interval. Parameters: $\mu = 0.1$, $\sigma = 0.5$, $p_0 = 3000$.

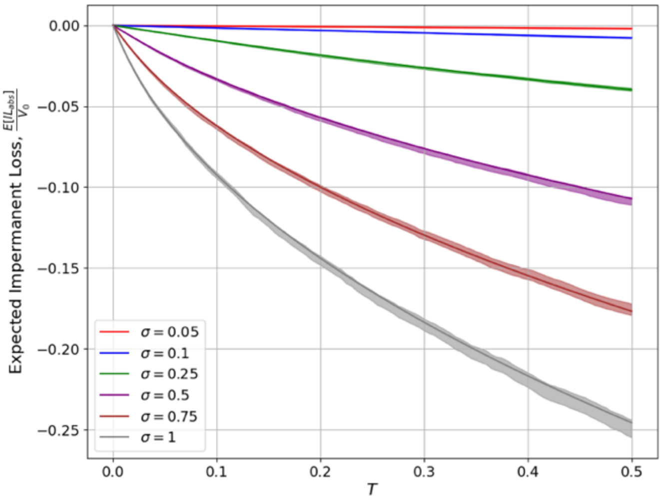

Figure 5. Expectation of IL as dependence of the investment horizon T for different volatility levels σ. Parameters: $\mu = 0$, $p_0 = 3000, p_l = 2500, p_u = 3000$.

3.2. Estimation of Expected Fee Income

We develop a framework for estimating fee income earned by LPs in Uniswap V3 concentrated liquidity pools. Our model incorporates realistic market microstructure while maintaining analytical tractability. We make the following key assumptions:

- **External Price Dynamics**: The external reference CEX price p_t follows a GBM (8) with approximately risk-neutral dynamics $(|\mu - \sigma^2/2| \approx 0)$ over the LP position's rebalancing horizon (e.g., hourly or daily).
- **Liquidity Position**: The LP provides concentrated liquidity within a fixed price range $[p_a, p_b]$. The position is active and earns fees if and only if the external price satisfies $p_t \in [p_a, p_b]$.
- **Fee Allocation**: Every swap, regardless of type, generates a fee equal to ϕ times its volume. Fees are collected by the LP only while the position is active.
- **Stationary Regime**: We analyse the system in its stationary state, where the distribution of $\frac{p_{\text{AMM}}(t)}{p_t}$ stabilises and the price does not fluctuate much. This is empirically justified for major Uniswap V3 pools, which typically reach equilibrium.
- **Transaction Flows**: The pool executes two types of swaps modelled by independent Poisson processes: ordinary swaps from end-users with intensity λ_{ord} and exogenous arbitrage that corrects price deviations. From Section 2.5, an arbitrage occurs when condition (10) is satisfied for $\phi > 0$.

These assumptions reflect the core mechanics of Uniswap v3 while abstracting from auxiliary effects (e.g., gas fee competition between arbitrageurs, liquidity shocks, position boundary effects and etc). Particularly, we introduce multiplicative constants C_{ord}, C_{arb}, which depend on market conditions, to mitigate these issues and allow empirical adjustments under this framework, the approximation of expected fee income can be estimated as follows.

Theorem 2. Consider a Uniswap V3 liquidity position providing liquidity within the price range $[p_l, p_u]$ with liquidity parameter $L_{pos} > 0$ and fee tier $\phi \in (0,1)$. Under the modelling assumptions above, the approximation of expected total fee income earned by the LP $E[\Phi(T)]$ over the time horizon $[0, T]$ is given by:

$$E[\Phi] \approx \phi(C_{ord}\overline{V_{\text{ord}}}\lambda_{\text{ord}} + C_{arb}\overline{V_{\text{arb}}}\lambda_{\text{arb}}) \cdot \frac{L_{\text{pos}}}{L_{\text{pool}}} \int_0^T P(p_t \in [p_a, p_b])\, dt,$$

where

- $C_{ord,arb}$ – calibration values which describe the model imperfections: C_{ord} captures the correlation between swap volumes and pool activity (e.g., larger trades occur when activity is high) and C_{arb} additionally adjusts for incomplete arbitrage execution (e.g., gas wars, failed transactions).
- $\overline{V_{\text{ord}}} = E[|V^{\text{ord}}|]$ is the expected absolute volume of an ordinary swap, $\overline{V_{\text{ord}}^2} = E[V_{\text{ord}}^2]$ – second-order moment of swap volume distribution.
- p_{avg} (L_{pool}) is a representative average price (liquidity of AMM) over the time interval $[0, T]$
- $\overline{V_{\text{arb}}} = \frac{1}{2} \cdot L_{pool} \cdot \sqrt{p_{\text{avg}}} \cdot (\phi + \epsilon)$ is the expected volume of an arbitrage swap.
- $\lambda_{\text{arb}} = \frac{\sigma^2 + \kappa\lambda_{\text{ord}}}{(\phi+\epsilon)^2}$ – the estimated intensity of arbitrage activity with $\kappa = \frac{4(1-\phi)^2\overline{V_{\text{ord}}^2}}{L_{\text{pool}}^2 \cdot p_{\text{avg.}}}$
- $P(p_t \in [p_a, p_b])$ is the probability that the external price p_t lies within the LP's active range at time t. The integral $\int_0^T P(p_t \in [p_a, p_b])\, dt$ gives the expected active lifetime of the position.

The first term in the theorem represents the expected fee income generated by ordinary trades $E[\Phi_{ord}]$, while the second term represents arbitrage-driven income $E[\Phi_{arb}]$.

Proof: See Appendix B.

To statistically analyse the proposed approach, we examined the impact of the constants C_{ord} and C_{arb} in Monte-Carlo simulations across a random set of market parameters, applying regression analysis. The details of the parameter analysis are provided in Appendix C, which enables us to understand the influence of individual macro- and microeconomic factors on the proportionality constants associated with different fee-generation mechanisms (ordinary and arbitrage transactions) in Theorem 2. The functional forms $C_{ord,arb} = f_{ord,arb}(\phi, \sigma^2, \dots)$ can be calibrated via Monte-Carlo methodology (Appendix C) or estimated directly from historical on-chain data via the least-squares procedure described in Appendix D. The empirical calibration results, sensitivity analysis, and out-of-sample validation for the ETH/USDC pool are presented in Section 4. However, the authors leave a detailed study of the dependencies of various factors on C_{ord} and C_{arb} for future research.

Figure 6 shows the profitability of the arbitrage component of fee income as computed using the proposed modelling framework. The fee parameter ϕ has a significant effect because it exhibits non-monotonic behaviour, i.e., for medium volatility ($\sigma = 0.5$), the expected profitability of arbitrage transactions increases as fees rise from 0% to approximately 1%, after which it begins to decline. Simulation results show that, for each level of volatility, there exists an optimal fee rate that maximises the expected return of a concentrated liquidity position. However, it is important to take into consideration that this experiment assumes a constant intensity of ordinary swaps – an assumption that does not hold in the real world, since fee levels directly influence the attractiveness of the pool for non-arbitrage transactions.

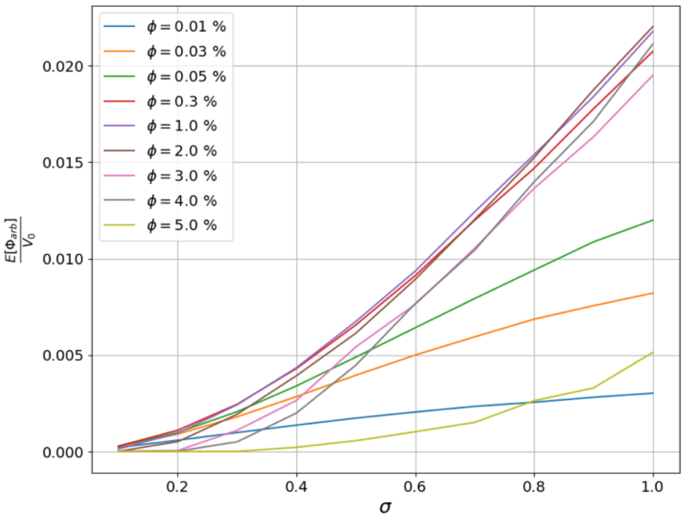

Figure 6. The impact of fee rate ϕ and volatility σ on expected fee income $\frac{E[\Phi_{arb}]}{V_0}$ generated by arbitrage transactions. Position parameters: initial value $V_0 = 10000$, geometrically symmetric half-width $\delta = 0.05$. Simulation parameters: $\lambda_{ord} = 1000, T = 1\ day, \mu = 0.0, \epsilon = 0,$ $E[|V_{ord}|] = 100, Var[V_{ord}] = 100,$ $p_0 = 3000, L_{pool} = 100 \cdot L_{pos}.$

The dependence of total fee income (from ordinary and arbitrage swaps) on position width in the zero-drift case ($\mu = 0$) is demonstrated in Figure 7. As the position width increase, the expected fee income decreases. Another key observation is that higher volatility leads to higher profitability, which aligns with empirical practice: greater volatility generates more arbitrage-driven fee income.

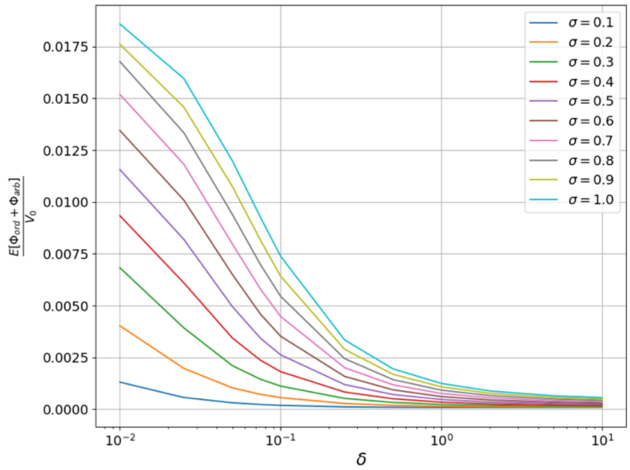

Figure 7. The dependence of total expected fee income $\frac{E[\Phi_{ord}+\Phi_{arb}]}{V_0}$ on the half-width $\delta = \frac{p_u}{p_0} = \frac{p_0}{p_l}$ of geometrically symmetric position under different levels of volatility σ. Position parameters: initial value $V_0 = 10000$. Simulation parameters: $\lambda_{ord} = 1000, T = 1\ day, \mu = 0.0, \epsilon = 0,$ $E[|V_{ord}|] = 100, Var[V_{ord}] = 100^2,$ $p_0 = 3000, L_{pool} = 100 \cdot L_{pos}, \phi = 0.05\%.$

3.3. Impermanent Profit

Although higher volatility increases the expected fee income, Figure 5 reveals that it simultaneously leads to larger expected ILes. This trade-off is examined in detail in Figure 8, which plots the expected LP profit:

$$IP = IL_{\text{abs}} + \Phi,$$

the sum of absolute IL and earned fees, as a function of position width for various volatility levels. Note that IP measures gross profitability before operational costs; for active strategies involving periodic rebalancing, the gas-adjusted net profit $P_{net}(T)$ defined in Section 4 provides a more accurate measure of realised returns. The dependence on volatility is non-monotonic: for example, at a narrow half-width $\delta = 0.1$, expected profit rises as σ increases from 0.1 to 0.7, but decreases for $\sigma > 0.7$. In contrast, position width exerts a monotonic influence on payoff across all volatility regimes – wider positions consistently yield lower (or at best equal) returns.

A second key observation is that, under the considered parameter settings, the expected value of a concentrated liquidity position in Uniswap V3 outperforms Uniswap V2 position with $\delta \to \infty$ when $\sigma \leq 1$, while the opposite holds for $\sigma \geq 1$. This illustrates an important property of the model: there exist market conditions (high volatility, low trade intensity or low fee tier), under which providing liquidity via Uniswap V2 is more profitable than using concentrated positions in Uniswap V3 on average. The authors consider a more detailed investigation of these conditions a promising direction for future work and a practically relevant problem for intelligent portfolio management in decentralised finance.

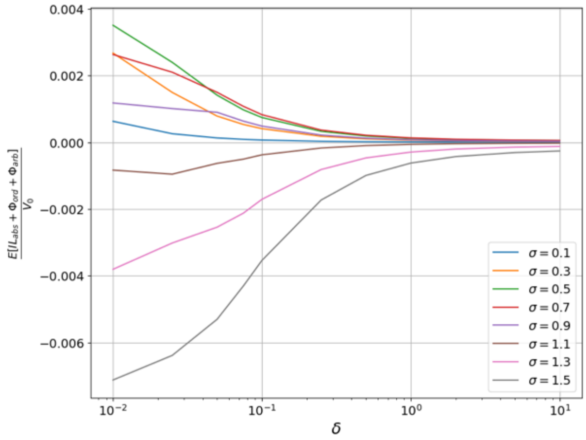

Figure 8. The dependence of Impermanent Profit IP (sum of Impermanent Loss IL and Fee Generation Φ) on the half-width of a geometrically symmetric position δ under different levels of volatility σ. Simulation parameters: $\lambda_{ord} = 1000$, $T = 1\ day, \mu = 0.0, \epsilon = 0, E[|V_{ord}|] = 500, Var[V_{ord}] = 1000^2, p_0 = 3000, L_{pool} = 10^7, V0 = 10^4, \phi = 0.05\%$.

We now examine the dynamics of the LP profit IP under two qualitatively distinct volatility regimes (low and high) across different liquidity ranges $[p_l, p_u]$. Figure 9 and Figure 10 show the dependence of IP on the position bounds for $\sigma = 0.5$ and $\sigma = 1.3$, respectively. For low volatility $\sigma = 0.5$ among all positions with a fixed width $\Delta = p_u - p_l$, there exists an optimal position range $[p_l^*, p_u^*]$ that maximises IP. In contrast, for the high volatility case $\sigma = 1.3$, no such optimum exists and the expected profit is maximised by the widest possible position ($p_l \to 0$ and $p_u \to \infty$), which corresponds to an Uniswap V2-style position. It is also worth noting that the optimal range $[p_l^*, p_u^*]$ for a given fixed width exhibits a strictly nonlinear dependence on market parameters, reflecting the complex interconnection between price dynamics, IL and fee income.

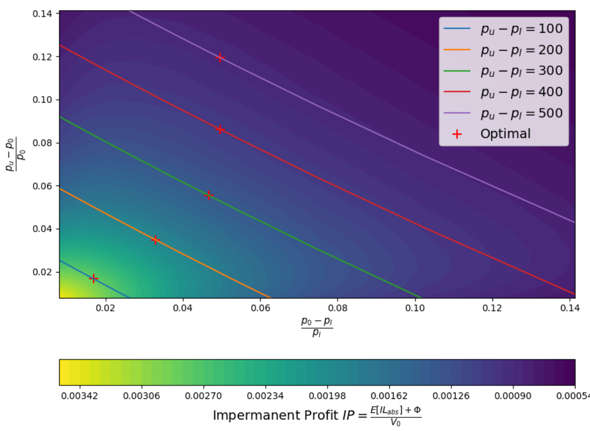

Figure 9. The impact of different position ranges $[p_l, p_u]$ on the impermanent profit at fixed volatility $\sigma = 0.5$. Simulation parameters: $\lambda_{ord} = 1000, T = 1\ day, \mu = 0.0, \epsilon = 0$, $E[|V_{ord}|] = 500, Var[V_{ord}] = 1000^2, p_0 = 3000, L_{pool} = 10^7, V0 = 10^4, \phi = 0.05\%$.

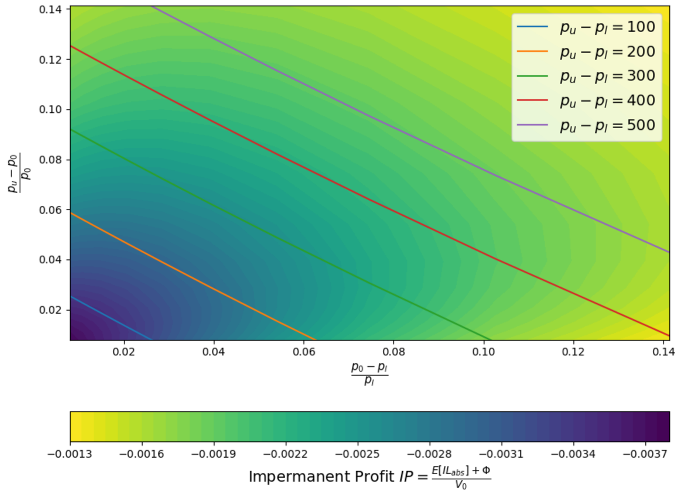

Figure 10. The impact of different position ranges $[p_l, p_u]$ on the impermanent profit at fixed volatility $\sigma = 0.5$. Simulation parameters: $\sigma = 1.3, \lambda_{ord} = 1000, T = 1\ day, \mu = 0.0, \epsilon = 0, E[|V_{ord}|] = 500, Var[V_{ord}] = 1000^2, p_0 = 3000$, $L_{pool} = 10^7, V0 = 10^4, \phi = 0.05\%$.

4. Empirical Results

To validate our stochastic framework, we parsed Ethereum data from 21000000 (19 Oct, 2024) to 23500000 (3 Oct, 2025) block numbers for a popular pool (ETH/USDC, 0.05% fee tier), capturing all swap transactions, liquidity adjustments and price oracle updates. As a practical application of our theoretical model, we analyse the performance of 1-day active rebalancing strategies for concentrated liquidity positions illustrated in Figure 11. Key model parameters for the simulation and calibration constants were quantified based on weekly swap dynamic history.

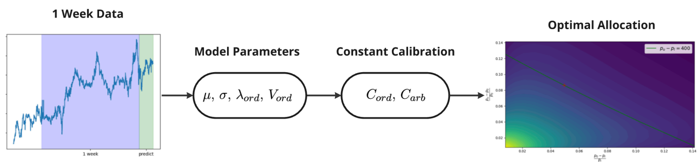

Figure 11. Concentrated Position Rebalancing Pipeline.

As one of the performance metrics, we use the annualised percentage yield (APY) defined as

$$APY_T = 365 \cdot APD_T = 365 \cdot \frac{V_T - V_{T-1}}{V_T},$$

which estimates the annualised return based on the change in the value from V_{T-1} to V_T of an active liquidity position over a single day T. The second metric is the Sharpe ratio, which captures the cumulative effect of reinvestment and is defined as

$$\text{Sharpe Ratio} = \frac{E_t[APY_t]}{\sigma_t(APY_t)},$$

i.e., the ratio of the expected daily APY to its standard deviation. This metric favours strategies that deliver high average returns while exhibiting low variability – thus reflecting both profitability and risk-adjusted stability. The rebalancing results are summarised in Table 1. Three families of strategies were considered: Uniswap V2, symmetric positions (half-widths of ±5, 15, 30%) and optimally placed positions (Figure 9). The worst performance was observed for the narrowest strategies (10% width), which suffer from high IL and, as a result, yield negative returns. Notably, Uniswap V2 serves as a strong baseline: simple symmetric rebalancing fails to outperform it. However, when positions are placed optimally, a significant improvement is achieved, yielding approximately 5-8% higher annualised APY compared to the corresponding symmetric strategy. Moreover, for the strategy with 60%-width configuration, both APY and Sharpe ratio exceed those of the Uniswap V2 strategy, emphasising the benefit of concentration when combined with optimal placement and demonstrating practical relevance of the proposed theoretical optimisation framework.

Table 1. Quantitative performance of the rebalancing strategies over the examined time period. Initial position value: $V_0 = 10000$

Position width $\frac{p_u - p_l}{p_0}$, %	Strategy	Final value $\frac{V_T}{V_0}$, %	Average APY, %	Sharpe ratio
Infinite	**Uniswap V2**	**132.87**	**34.94 ± 61.98**	**0.0567**
10%	**Symmetric** $P_0 \pm 5\%$	90.49	-1.99 ± 74.55	-0.0026
	Optimal	**91.88**	**3.75 ± 72.23**	**0.0005**
30%	**Symmetric** $P_0 \pm 15\%$	120.19	25.11 ± 63.74	0.0387
	Optimal	**126.49**	**30.35 ± 59.21**	**0.0510**
60%	**Symmetric** $P_0 \pm 30\%$	126.30	29.06 ± 57.94	0.0502
	Optimal	**136.21**	**37.13 ± 60.21**	**0.0612**

It should be noted that the proposed approach does not account for blockchain network fees incurred with each rebalancing operation. Moreover, in practice, LP often employ trigger-based rebalancing strategies, where positions are rebalanced not at fixed time intervals, but when a certain condition is met. Such rule-based approaches can reduce risk of IL and increase returns by avoiding unnecessary transactions. Extending the current framework to analyse these adaptive, constraint-driven strategies lies beyond the scope of current study. Nevertheless, the authors regard this as a challenging and highly relevant direction for future research, with direct practical value for DeFi investors and Automated Liquidity Management systems.

5. Limitations

While the proposed framework provides a tractable analytical model for evaluating liquidity provision in Uniswap V3 under stochastic price dynamics, several simplifying assumptions require discussion. These limitations reflect trade-offs between realism and mathematical feasibility, and they point toward promising directions for future work. Next, let us discuss the main points of these imperfections.

Our theoretical approach considers that arbitrageurs act instantaneously and fully eliminate price deviations once the no-arbitrage band is intersected. In practice, on-chain arbitrage is subject to execution delays, gas wars, and strategic behaviour (e.g., front-running, back-running or sandwich attacks). These frictions reduce arbitrage efficiency and may lead to persistent price deviations. While our constants C_{arb} and C_{ord} can be calibrated on historical data to absorb some of these effects, a more granular and complex model incorporating miner extractable value (MEV) [24] dynamics and arbitrageurs' rationality would improve analysis. Moreover, the theoretical model explored in this work assumes independence between arbitrage and ordinary swaps, whereas in practice arbitrage activity often follows large ordinary trades. In our theoretical derivations, we adopt this simplifying assumption to obtain closed-form expressions that are interpretable.

The model assumes a stationary regime where pool liquidity L_{pool}, GBM parameters (drift μ, volatility σ), and swap intensities (λ_{ord}, λ_{arb}) remain constant over the investment horizon. However, real-world DeFi pools have time-varying liquidity, dynamics that depend on news or external market conditions, and non-Poisson trade arrivals. Extending the framework to time-inhomogeneous and more flexible stochastic processes like Merton jump-diffusion or Heston process would better capture these dynamics.

6. Conclusions

This work establishes a unified stochastic framework for valuing concentrated liquidity positions in Uniswap V3, bridging theoretical price dynamics, arbitrage mechanics and empirical market microstructure.

Our core contributions demonstrate that liquidity provision is inherently contextual. Theorem 1 provides the first closed-form solution for the expected IL of concentrated positions, revealing its nonlinear scaling with volatility, time horizon and position range width. Theorem 2 decomposes fee income into ordinary and arbitrage-driven components, explicitly linking arbitrage intensity to the variance-balance mechanism. Crucially, our model identifies a volatility-dependent profitability threshold: as shown in the simulation in Figure 8, Uniswap V2's uniform liquidity outperforms narrow Uniswap V3 positions due to excessive IL, while at lower volatility concentrated positions with optimally placed ranges can generate higher returns. This challenges the prevailing heuristic that "narrower ranges always maximise fees" and provides a quantitative foundation for adaptive liquidity management. The empirical results indicate that the conventional practice of opening positions with fixed, symmetric ranges (±5%, ±15%, ±30%) can be suboptimal. The proposed framework provides a procedure for optimising expected returns.

The practical applicability of our model requires for further incorporation of operational costs, which can substantially shift the optimal position width. In networks with high transaction fees, such as Ethereum, it is economically rational to use wider price ranges and less frequent rebalancing. In Uniswap V3 during high volatility periods, which often occur in crisis market conditions, the priority for LP shifts from fee maximisation to minimising IL, making the unbounded liquidity of V2 a more profitable strategy.

The current research describes Uniswap-specific mechanics to establish a foundational methodology for rational capital allocation in decentralised markets. By quantifying the trade-offs between fee income and IL, we empower LP to systematically estimate the risks-return landscape and protocol designers to align fee rates with sustainable liquidity provision to profit maximisation. As concentrated liquidity becomes the de-facto standard for AMMs, this work provided both a theoretical compass over Uniswap V3 and a yet another practical toolkit for AMM analysis.

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None declared.

Ethical Approval:
Not applicable.

Author's Contribution:
All authors designed and coordinated this research and prepared the manuscript in entirety.

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Appendix A. Proof of Theorem 1

A.1 Truncated k-th Moment of Log-Normal Distribution

Lemma. Let z is log-normal random variable, i.e., $\ln z \sim \mathcal{N}(\mu, \sigma^2)$. Then the truncated k-th moment is given by

$$\int_0^{z_u} z^k f_z(z)\, dz = e^{k\mu+\frac{1}{2}k^2\sigma^2}\Phi\left(\frac{\ln z_u - \mu - k\sigma^2}{\sigma}\right),$$

where $f_z(z)$ is the probability density function of z and $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution $\mathcal{N}(0,1)$.

Proof. Using the substitution $z = e^x$, where $x \sim \mathcal{N}(\mu, \sigma^2)$, we obtain

$$\int_0^{z_u} z^k f_z(z)\, dz = \int_{-\infty}^{\ln z_u} e^{kx} \cdot \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx =$$

$$= e^{k\mu+\frac{k^2\sigma^2}{2}} \int_{-\infty}^{\ln z_u} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{(x-\mu-k\sigma^2)^2}{2\sigma^2}\right) dx =$$

$$= e^{k\mu+\frac{1}{2}k^2\sigma^2}\Phi\left(\frac{\ln z_u - \mu - k\sigma^2}{\sigma}\right),$$

which yields the desired expression.

A.2 Properties of GBM Dynamics

From the properties of GBM for p_t, we know that the parameters of the log-normal distribution are

$$\mu_{p,t} = \ln p_0 + \left(\mu - \frac{\sigma^2}{2}\right)t,$$
$$\sigma_{p,t}^2 = \sigma^2 t.$$

We now apply the previous lemma to obtain the following expressions when evaluating the integrals:

$$\int_a^b f_{p_t}(p)\, dp = \Phi(d_1(b)) - \Phi(d_1(a)),$$

$$\int_a^b p_t\, f_{p_t}(p) dp = p_0 e^{\mu t}[\Phi(d_2(b)) - \Phi(d_2(a))],$$

$$\int_a^b \sqrt{p}\, f_{p_t}(p) dp = \sqrt{p_0} e^{\left(\frac{\mu}{2}-\frac{\sigma^2}{8}\right)t}[\Phi(d_3(b)) - \Phi(d_3(a))],$$

where

$$d_1(x) = \frac{\ln\left(\frac{x}{p_0}\right) - \left(\mu - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}},$$

$$d_2(x) = d_1(x) - \sigma\sqrt{t},$$

$$d_3(x) = d_1(x) - \frac{\sigma\sqrt{t}}{2}.$$

A.3. Expectation of Position Value

Using [V3], we decompose the expected value of the concentrated liquidity position as follows:

$$E_{p_t}[V_y(p_t)] = \underbrace{\int_0^{p_l} V_y(p) f_{p_t}(p) dp}_{E_1: p_t \le p_l} + \underbrace{\int_{p_l}^{p_u} V_y(p) f_{p_t}(p) dp}_{E_2: p_l < p_t < p_u} + \underbrace{\int_{p_u}^{\infty} V_y(p) f_{p_t}(p)\, dp}_{E_3: p \ge p_u}.$$

Let us consider E_1 and E_3 terms using A.2 and explicit form of $V_y(p)$ from [V3]:

$$E_1 = L\left(\frac{1}{\sqrt{p_l}} - \frac{1}{\sqrt{p_u}}\right) \cdot p_0 e^{\mu t} \cdot \Phi\left(\frac{\ln\frac{p_l}{p_0} - \left(\mu + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right),$$

$$E_3 = L(\sqrt{p_u} - \sqrt{p_l})\left[1 - \Phi\left(\frac{\ln\frac{p_u}{p_0} - \left(\mu - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right)\right].$$

Considering the case where p_t lies within the position's price range, we write:

$$E_2 = 2L\int_{p_l}^{p_u} \sqrt{p}\, f_{p_t}(p) dp - L\sqrt{p_l}\int_{p_l}^{p_u} f_{p_t}(p)\, dp - \frac{L}{\sqrt{p_u}}\int_{p_l}^{p_u} p\, f_{p_t}(p) dp.$$

Substituting the result from A.2 into each integral sequentially, we obtain:

$$E_2 = 2L\sqrt{p_0}e^{\left(\frac{\mu}{2}-\frac{\sigma^2}{8}\right)t}\left[\Phi\left(\frac{\ln\frac{p_u}{p_0} - \mu t}{\sigma\sqrt{t}}\right) - \Phi\left(\frac{\ln\frac{p_l}{p_0} - \mu t}{\sigma\sqrt{t}}\right)\right] -$$

$$- L\sqrt{p_l}\left[\Phi\left(\frac{\ln\frac{p_u}{p_0} - \left(\mu - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right) - \Phi\left(\frac{\ln\frac{p_l}{p_0} - \left(\mu - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right)\right] -$$

$$-\frac{L}{\sqrt{p_u}} \cdot \mathrm{p}_0 e^{\mu t}\left[\Phi\left(\frac{\ln\frac{p_u}{p_0}-\left(\mu+\frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right)-\Phi\left(\frac{\ln\frac{p_l}{p_0}-\left(\mu+\frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right)\right].$$

Summing the terms E_1, E_2, and E_3 divided by $2L\sqrt{p_0}$ yields the desired expression. The theorem is proved.

Appendix B: Proof of Theorem 2

Let us define:

- N_t^{ord}: Ordinary swap process (Poisson with intensity λ_{ord}); $\{\tau_i^{\mathrm{ord}}\}$: Timestamps of trades; V_i^{ord}: Swap volumes, i.i.d.
- N_t^{arb}: Arbitrage swap process (Poisson with intensity λ_{arb}) $\{\tau_i^{\mathrm{arb}}\}$: Timestamps of trades; V_j^{arb}: Swap volumes, i.i.d.
- $I_t = I\{p_t \in [p_a, p_b]\}$: Position active indicator.

B.1 Fee Allocation Mechanism in Uniswap v3

In Uniswap v3, fees are distributed proportionally to liquidity share. When a swap of volume V occurs at price $p_t \in [p_a, p_b]$, the LP's fee share is

$$\mathrm{Fee}_{\mathrm{LP}} = \phi \cdot |V| \cdot \frac{L_{\mathrm{pos}}}{L_{\mathrm{pool}}(t)},$$

which holds for both ordinary and arbitrage swaps. The total fee income is

$$\Phi(T) = \phi \sum_{i=1}^{N_T^{\mathrm{ord}}} |V_i^{\mathrm{ord}}| \cdot I_{\tau_i^{\mathrm{ord}}} \cdot \frac{L_{\mathrm{pos}}}{L_{\mathrm{pool}}(\tau_i^{\mathrm{ord}})} + \phi \sum_{j=1}^{N_T^{\mathrm{arb}}} |V_j^{\mathrm{arb}}| \cdot I_{\tau_j^{\mathrm{arb}}} \cdot \frac{L_{\mathrm{pos}}}{L_{\mathrm{pool}}(\tau_j^{\mathrm{arb}})}.$$

B.2 Expected Fee Income Calculation

Taking expectations:

$$E[\Phi(T)] = \phi L_{\mathrm{pos}} \cdot E\left[\sum_{i=1}^{N_T^{\mathrm{ord}}} \frac{|V_i^{\mathrm{ord}}| I_{\tau_i^{\mathrm{ord}}}}{L_{\mathrm{pool}}(\tau_i^{\mathrm{ord}})}\right] + \phi L_{\mathrm{pos}} \cdot E\left[\sum_{j=1}^{N_T^{\mathrm{arb}}} \frac{|V_j^{\mathrm{arb}}| I_{\tau_j^{\mathrm{arb}}}}{L_{\mathrm{pool}}(\tau_j^{\mathrm{arb}})}\right].$$

Assuming $L_{\mathrm{pool}}(t)$ varies slowly compared to swap arrivals (justified empirically for major pools), we can approximate with average pool liquidity as $L_{\mathrm{pool}}(\tau_i) \approx L_{\mathrm{pool}}$. Then, using independence of V_i^{ord} and the estimation for Poisson process $N_T^{ord} \approx \lambda_{\mathrm{ord}} T$, we can rewrite:

$$E\left[\sum_{i=1}^{N_T^{\mathrm{ord}}} \frac{|V_i^{\mathrm{ord}}| I_{\tau_i^{\mathrm{ord}}}}{L_{\mathrm{pool}}(\tau_i^{\mathrm{ord}})}\right] \approx \mathrm{C}_{\mathrm{ord}} \cdot \overline{V_{\mathrm{ord}}} \cdot \lambda_{\mathrm{ord}} \cdot \frac{L_{\mathrm{pos}}}{L_{\mathrm{pool}}} \int_0^T P(p_t \in [p_a, p_b])\, dt,$$

and similarly for arbitrage swaps:

$$E\left[\sum_{i=1}^{N_T^{\mathrm{arb}}} \frac{|V_i^{\mathrm{arb}}| I_{\tau_i^{\mathrm{arb}}}}{L_{\mathrm{pool}}(\tau_i^{\mathrm{arb}})}\right] \approx \mathrm{C}_{\mathrm{arb}} \cdot \overline{V_{\mathrm{arb}}} \cdot \lambda_{\mathrm{arb}} \cdot \frac{L_{\mathrm{pos}}}{L_{\mathrm{pool}}} \int_0^T P(p_t \in [p_a, p_b])\, dt,$$

where C_{ord}, C_{arb} – calibration values, which introduces to mitigate model assumption imperfections. Combining these terms:

$$E[\Phi(T)] \approx \phi(C_{ord}\overline{V_{\mathrm{ord}}}\lambda_{\mathrm{ord}} + C_{arb}\overline{V_{\mathrm{arb}}}\lambda_{\mathrm{arb}}) \cdot \frac{L_{\mathrm{pos}}}{L_{\mathrm{pool}}} \int_0^T P(p_t \in [p_a, p_b])\, dt.$$

Note that if the price dynamics and the Poisson process of regular transactions are independent, then the constant $C_{ord} = 1$ holds exactly, without any approximations.

The integral in the expressions of estimated income, using GBM dynamics, can be rewritten as

$$\int_0^T P(p_t \in [p_a, p_b])\, dt = \int_0^T \left(\Phi\left(\frac{\ln\frac{p_u}{p_0}-\left(\mu-\frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right)-\Phi\left(\frac{\ln\frac{p_l}{p_0}-\left(\mu-\frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right)\right) dt.$$

B.3 Estimation of $\overline{V_{\mathrm{arb}}}$

From Uniswap v3 mechanics, the volume required to move the price from $p_{\mathrm{before}} = p_t(1 + \phi + \epsilon)$ to $p_{\mathrm{after}} \approx p_t$, then, from (3),

$$|V^{\mathrm{arb}}| = L_{\mathrm{pool}}(t)\sqrt{p_t} \cdot \left|\sqrt{1 + \phi + \epsilon} - 1\right|.$$

Using $\sqrt{1+x} \approx 1 + x/2$:

$$\overline{V_{\mathrm{arb}}} = E\left[|V^{\mathrm{arb}}|\right] = \frac{1}{2} \cdot L_{\mathrm{pool}}(t) \cdot \sqrt{p_{\mathrm{avg}}} \cdot (\phi + \epsilon).$$

B.4 Impact of Ordinary Swaps

Each ordinary swap of volume V_i^{ord} induces a price change $p(\tau_i) \to p(\tau_{i+1})$ in the AMM. From Uniswap V3's core equation:

$$\sqrt{p(\tau_{i+1})} - \sqrt{p(\tau_i)} = \frac{(1-\phi)V_i^{ord}}{L_{pool}(\tau_i)}.$$

For small deviations, the change for logarithmic price $Z_t^{ord}(\tau_i) = \ln\left(\frac{p(\tau_i)}{p_t}\right)$ due to ordinary swap, using $\ln(1+x) \approx 1 + x$, is estimated as

$$\Delta Z_t^{ord}(\tau_i) = Z_t(\tau_{i+1}) - Z_t(\tau_i) = 2\ln\left(1 + \frac{(1-\phi)V_i^{ord}}{L_{pool}(\tau_i)\sqrt{p(\tau_i)}}\right) \approx \frac{2(1-\phi)V_i^{ord}}{L_{\mathrm{pool}} \cdot \sqrt{p_{\mathrm{avg}}}}$$

Let $\xi_i = \Delta Z_t^{ord}(\tau_i)$ at the i-th swap. Then

$$E[\xi_i] \approx 0, \qquad Var(\xi_i) = \kappa \approx \frac{4(1-\phi)^2\overline{V_{\mathrm{ord}}^2}}{L_{\mathrm{pool}}^2 \cdot p_{\mathrm{avg}}}.$$

B.5 Risk-Neutral Dynamic of Price Deviation

Under the assumption of $|\mu - \sigma^2/2| \approx 0$, the external price follows

$$d\ln p_t \approx \sigma dW_t.$$

Between swap events, the AMM price $p_{\mathrm{AMM}}(t)$ remains constant, so the deviation process $Z_t = \ln\left(\frac{p_{AMM}(t)}{p_t}\right)$ evolves as

$$dZ_t = -d\ln p_t \approx -\sigma dW_t.$$

B.6 Variance-Balance Equation

The total rate of variance accumulation for Z_t has two components: continuous diffusion from market volatility (σ^2) and discrete jumps from ordinary independent swaps ($\lambda_{\mathrm{ord}} \cdot \kappa$). The total variance growth rate is:

$$\Gamma_{\mathrm{growth}} = \sigma^2 + \lambda_{\mathrm{ord}}\kappa.$$

Arbitrageurs act when $|Z_t| > a$, where $a = \phi + \epsilon$ is the no-arbitrage threshold derived in Section 2.5. Therefore, the variance reduction rate due to arbitrage swap can be estimated as:

$$\Gamma_{\text{reduction}} = \mathrm{d}_{\text{arb}} \cdot \lambda_{\text{arb}} \cdot a^2 = d_{arb} \cdot \lambda_{\text{arb}} \cdot (\phi + \epsilon)^2,$$

where d_{arb} – the constant of arbitrage variance reduction effectivity. However, one can note, that d_{arb} is absorbed into the calibration constant C_{arb} in Theorem 2, therefore, we could derive further equations with assumption of $d_{arb} = 1$.

In the stationary regime, the rate of variance growth equals the rate of variance reduction, using market microstructure,

$$\Gamma_{\text{growth}} = \Gamma_{\text{reduction}}.$$

Substituting the components, we could formulate variance-balance equation:

$$\sigma^2 + \lambda_{\text{ord}}\kappa = \lambda_{arb} \cdot (\phi + \epsilon)^2.$$

B.7 Estimation of λ_{arb}

Solving previous equation in terms of λ_{arb}:

$$\lambda_{\text{arb}} = \frac{\sigma^2 + \kappa\lambda_{\text{ord}}}{(\phi + \epsilon)^2} = \frac{\sigma^2 + 4(1-\phi)^2 \frac{\lambda_{\text{ord}} \overline{V_{\text{ord}}^2}}{L_{\text{pool}}^2 \cdot p_{\text{avg}}}}{(\phi + \epsilon)^2}.$$

This completes the derivation of key components for Theorem 2. The final expression is obtained by combining Sections B.2-B.7.

Appendix C: Regression Analysis of C_{ord} and C_{arb}

As features, the set S_i of parameters satisfying the next specified conditions were used:

$$0.0001 \leq \frac{L_{pos}}{L_{pool}} \leq 0.1,$$

$$1000 \leq L_{pool} \leq 1000000,$$

$$-0.4 \leq \mu \leq 0.4,$$

$$1 \leq \sigma^2 \leq 1,$$

$$100 \leq \lambda_{ord}(per\ day) \leq 2000,$$

$$0 \leq \phi \leq 0.01,$$

$$0 \leq p_l \leq p_0 \leq p_u \leq 20000,$$

$$100 \leq Var[V_{ord}] \leq 1000,$$

and $p_0 = 3000$, $\epsilon = 0$, $T = 1\,day$, $E[|V_{ord}|] = 500$. To simulate the stochastic arrival of ordinary users, a log-normal distribution for $|V_{ord}|$ with fixed $E[|V_{ord}|]$ and $Var[|V_{ord}|]$ is employed.

For each parameter set, Monte-Carlo simulations with $N_{sim} = 5000$ are used to estimate the expected returns from both ordinary and arbitrage transactions, enabling the calibration of the proportionality constants C_{arb} and C_{ord} as defined in Theorem 2. The regression analysis is performed on a sample of $N = 3{,}000$ observations $\left(f(S_i), C_{ord,arb}\right)$, where the features $f(S_i)$ are the corresponding dimensionless quantities listed in Table 2 and Table 3. The intercept is statistically significant for C_{ord}, while all other considered parameters are insignificant at the $\alpha = 0.05$ (confidence level). The obtained result ($C_{\text{ord}} \approx 1$) is fully consistent with the theoretical prediction (see Appendix B.2), which based on model's assumption that the Poisson order arrival process is independent of the underlying GBM price dynamics. The most significant predictors for C_{arb} are the intercept, the fee level ϕ (pool parameter), the GBM volatility σ, and the variance of ordinary swap volumes $V[V_{ord}]$. Notably, the position width parameters $[p_l, p_u]$ were found to be statistically insignificant. This observation provides a basis for further optimising the position width maximising expected returns, while holding C_{ord} fixed under given market conditions.

Table 2. Coefficients and corresponding p-values, indicating the statistical dependencies of the considered parameters on C_{ord} estimated via Monte-Carlo sampling

Parameters	coefficient	p-value	Significant
intercept	0.999	≈ 0	+
μT	-0.202	0.665	–
$\sigma^2 T$	-0.306	0.674	–
$\frac{p_l}{p_0}$	≈ 0	0.786	–
$\frac{p_u}{p_0}$	≈ 0	0.273	–
$\frac{L_{pos}}{L_{pool}}$	0.002	0.832	–
$\frac{\sqrt{V[V_{ord}]}}{V_0}$	≈ 0	0.465	–
$\lambda_{ord}T$	≈ 0	0.384	–
ϕ	0.119	0.242	–

Table 3. Coefficients and corresponding p-values, indicating the statistical dependencies of the considered parameters on C_{arb} estimated via Monte-Carlo sampling.

Parameters	coefficient	p-value	Significant
intercept	0.825	≈ 0	+
μT	2.517	0.522	–
$\sigma^2 T$	-20.486	≈ 0	+
$\frac{p_l}{p_0}$	-0.010	0.280	–
$\frac{p_u}{p_0}$	-0.001	0.489	–
$\frac{L_{pos}}{L_{pool}}$	0.024	0.776	–
$\frac{\sqrt{V[V_{ord}]}}{V_0}$	-0.004	0.351	–
$\lambda_{ord}T$	≈ 0	0.685	–
ϕ	24.331	≈ 0	+